Second Scientific Report

"Curso de Ciências Moleculares da Universidade de São Paulo"

Condensed matter analogs of the quantum vacuum gravitational awakening effect

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1 Developed Activities

This project's aim is to search for condensed matter analogs of the vacuum awakening effect (VAE) [1, 2, 3].

Having previously studied general relativity following [4] and the basics of condensed matter field theory using [5], I could, under supervision of prof. George E. A. Matsas, identify what could be a promising analog for VAE, more specifically, a system under phase transition possibly displaying tachyonic modes.

In the following, I first briefly describe what is the VAE, then I discuss how it can be understood from the perspective of phase transitions in condensed matter physics.

2 Vacuum Awakening

The Vacuum Awakening Effect (VAE) [1, 2, 3], first studied by W. C. C. Lima and D. A. T. Vanzella in 2010, predicts the exponentially fast increase of the vacuum energy density for a scalar field properly coupled to the spacetime curvature, induced by the formation of a relativistic massive body, like a static neutron star. This increase ("awakening") will overwhelm the energy density of the star, possibly destabilizing it and controlling its fate.

The description of the effect is given by a free scalar field of mass m non-minimally coupled to the Ricci scalar with coupling ξ:

$$\left(-\nabla^a\nabla_a + m^2 + \xi R\right)\phi = 0 \tag{1}$$

The background metric is taken to evolve from an asymptotically flat configuration in the past, to an asymptotically static and spherically symmetric configuration in the future, realized for example in the formation of a static relativistic star from collapsed low density matter:

$$ds^2 = \begin{cases} -dt^2 + dx^2 + dy^2 + dz^2 & \text{, past} \\ -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 & \text{, future} \end{cases}$$

where $d\Omega^2 = d\theta^2 + sin^2\theta d\varphi^2$.

A general solution to (1) will be an expansion in terms of modes: $\phi = \sum(a_k u_k + \tilde{a}_k u_k^*)$. The quantum field operator $\hat{\Phi}$ can be expanded using the same modes, but the coefficients a_k are converted to creation/annihilation operators that obey $[a_k, a_{k'}^\dagger] = 1$:

$$\hat{\Phi} = \sum_k \left[\hat{a}_k u_k + \hat{a}_k^\dagger u_k^*\right]$$

The vacuum state will be defined by $a_k|0\rangle = 0$, and our solutions u_k will dictate the behaviour of observables such as the field amplitude and energy-momentum tensor.

In the past, solutions to (1) will take the form of plane waves $u_k \sim e^{i(\omega_k t - \mathbf{k}\cdot x)}$. However, in the future, the general solution takes the (properly normalized) form

$$v_{\omega l \mu} \propto \frac{1}{r} e^{-i\omega t} F_{\omega l}(r) Y_{l\mu}(\theta, \varphi)$$

where $Y_{l\mu}$ are the spherical harmonics, and $F_{\omega l}$ are the radial part. Writing (1) in terms of metric components[1], we end up with an effective potential for F:

$$\left[-f\partial_r\left(f\partial_r\right) + V_{\text{eff}}^{(l)}\right] F_{\omega l} = \omega^2 F_{\omega l} \tag{2}$$

where

$$V_{\text{eff}}^{(l)} = f(r)\left(m^2 + \frac{l(l+1)}{r^2} + \xi R + \frac{\partial_r f(r)}{r}\right) \tag{3}$$

There are a handful of observations to be made here: we see that the in- and out-vacuum are not the same, so that particle creation must take place (which nevertheless is a negligible effect here). More importantly, negative eigenvalue $\omega^2 = -\Omega^2 < 0$ solutions to (2) have $\exp(\pm\Omega t)$ dependence:

$$w_{\Omega l \mu} \sim \left(e^{-\Omega t + i\pi/12} + e^{\Omega t - i\pi/12}\right) \frac{F_{\Omega l}(r)}{r} Y_{l\mu}(\theta, \varphi)$$

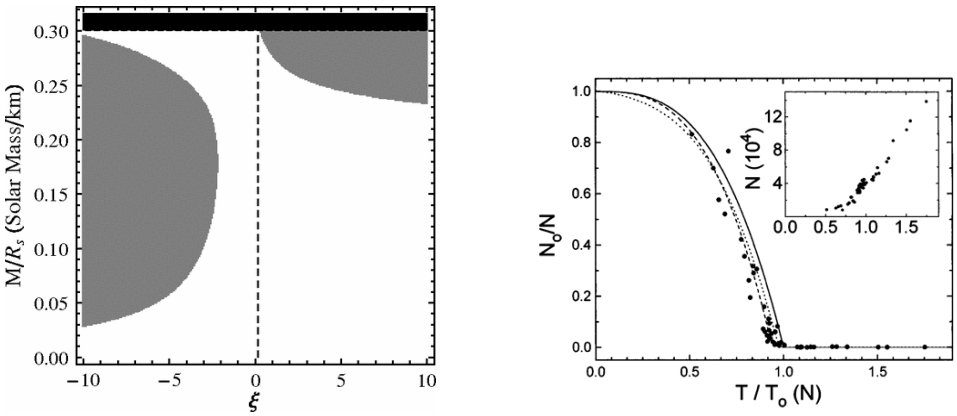
These are called tachyonic modes. Their existence implies that part of the plane-wave modes eventually undergo an exponential growth, leading to an unbounded increase of the vacuum expectation value of $\hat{\Phi}^2$.

In this case, at each point of spacetime, $\langle \hat{T}_{\mu\nu} \rangle$ will exponentially increase or decrease (conserving the net energy-momentum). In the case of no initial perturbations, this expectation value must be taken over the vacuum: $\langle 0|\hat{T}_{\mu\nu}|0\rangle$, that is, the vacuum energy dominates the spacetime evolution.

As for the fate of the star, not much can be said as this would require an analysis beyond the linear regime of the field, although for negative coupling ξ it has been suggested that the star would undergo "spontaneous scalarization", and the final state would be a static star, with a non zero scalar field [6], however it is possible that the vacuum awakening could lead to an implosion or explosion scenario, for $\xi > 0$.

Clearly, the presence of tachyonic modes depends on the capacity of V_{eff} to support negative eigenvalue solutions. Hence, we would like to explore the behaviour of this effective potential as parameters like the star's mass, radius and ξ are varied.

[1] We are using here that $\nabla^\mu \nabla_\mu = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu\right) = -f(r)^{-1}\partial_t^2 + \frac{1}{r^2}\partial_r\left(r^2 f(r)\partial_r\right) - \frac{\hat{L}^2}{r^2}$, where $g = \det(g_{\mu\nu})$, $g^{\mu\nu}$ is the inverse metric and $\hat{L}$ is the angular momentum operator



(a) (b)

Figure 1: This illustrates the kind of analogy that we pursue between VAE and condensed matter phase transitions. In (a), the shaded regions represent values of the ratio M/R_s between the mass of the star and its radius, and of the coupling constant ξ for which the VAE is triggered for a massless scalar field. The black region represents mass values for which no classical equilibrium is possible. (Reproduced from [2]). In (b), the formation of a Bose-Einstein condensate of ^{87}Rb atoms is displayed. This is a graph of the fraction of particles in the ground state (N_0/N) as a function of the scaled temperature in relation to the critical temperature T_0. Although not important for the discussion here, the solid (dotted) line represents the prediction from a theory of an infinite (finite) number of non interacting bosons, and the dashed line is the least-squares fit of the experimental points to the theoretical model. (Reproduced from [7]).

3 Vacuum Awakening as a Phase Transition

We must now ask ourselves: under which conditions will (2) admit negative eigenvalues? This of course will depend on the behaviour of the effective potential (3). We can take a massless field $m = 0$ with $l = 0$, as these are the values for which the VAE will be triggered most easily. The remaining factors involving R and $f(r)$ can be related to the matter distribution through Einstein's equations.

As shown in [2], taking a uniform density star, there are values of ξ and the mass-radius ratio of the star (figure 1a) for which the effect is triggered, and the vacuum energy density becomes dominant over the star density in a time scale as short as milliseconds.

This can be understood as a phase transition in a condensed matter system such as a magnet undergoing spontaneous symmetry breaking - the star's radius R_S acts as the temperature, and as it is varied, the system's behaviour changes significantly: the magnet acquires a permanent magnetization, and analogously, the vacuum of the scalar field is awakened.

Below, we describe the Ginzburg-Landau theory of phase transitions, and

other interesting examples of phase transition in condensed matter physics. First let us introduce some formalism.

3.1 The Coherent State Path Integral

In quantum mechanics, the Feynman path integral is used to calculate the amplitude of a single-particle coordinate q to change from a state q_I to q_F, by integrating over all paths in space-time connecting q_I to q_F, weighted by the exponential of their action:

$$\langle q_F|e^{-iHt}|q_I\rangle = \int Dq(t)e^{iS[q(t)]}$$

where H is the Hamiltonian of the particle. We wish to extend this idea to quantum field theory. For this, we need to give a meaning to an integration over field configurations. In the case of a single particle, this is done by performing a time slicing procedure where it is possible to approximate $\exp(-i\hat{H}(\hat{p},\hat{q})\Delta t)$ by $\exp(-iH(p_n,q_n)\Delta t)$ and integrate over all values of (q_n,p_n), the coordinate and momentum eigenstates at a time step n. Similarly, in condensed matter systems, where a Hamiltonian is usually specified in terms of creation and annihilation operators, $\hat{H} = H(\hat{a}^\dagger,\hat{a})$, the integration must be performed over all values of creation and annihilation eigenstates: the so-called *coherent states*.

3.1.1 Canonical Coherent State

For a harmonic oscillator, the coherent state is given by

$$|\phi\rangle := e^{\phi a^\dagger}|0\rangle$$

where $a^\dagger$ is the creation operator and ϕ is a complex number. An important property of $|\phi\rangle$ is that it is an eigenstate of the annihilation operator:

$$a|\phi\rangle = \phi|\phi\rangle$$

and similarly, $\langle\phi|a^\dagger = \langle\phi|\bar{\phi}$, where $\bar{\phi}$ denotes complex conjugation. It is easy to check that the number of particles in a coherent state is given by a Poisson distribution with average $\bar{\phi}\phi = |\phi|^2$

$$P(n) = \frac{\langle n|\phi\rangle^2}{\langle\phi|\phi\rangle} = \frac{(\bar{\phi}\phi)^n e^{-\bar{\phi}\phi}}{n!}$$

In fact, the coherent state is the state with minimum possible uncertainty. This means that, for a coherent state, we would always find

$$\Delta x\Delta p = \frac{\hbar}{2}$$

resembling the most a classical state. Indeed, the expectation value of $q = \frac{a+a^\dagger}{\sqrt{2}}$ executes simple harmonic motion, like in classical mechanics:

$$\langle \hat{q} \rangle = \sqrt{2}\phi cos(\omega t)$$

3.1.2 Many-body Coherent States

We can construct coherent states for many-body systems as well. We restrict our discussion to bosonic systems for simplicity. Consider the Hilbert space $\mathcal{H}$ for a single-particle system with basis of states $\{|1\rangle, |2\rangle, ...\}$. The state of a many-body system resides in a Fock space, which is the direct sum of symmetrized products of single-particle Hilbert spaces:

$$\mathcal{F} = \bigoplus_{N \geq 0} \mathrm{Sym}^N(\mathcal{H})$$

where Sym^N denotes the Nth symmetric power. We set $\mathrm{Sym}^0(\mathcal{H}) = \{|0\rangle\}$ and call $|0\rangle$ the vacuum state.

A state in $\mathrm{Sym}^N(\mathcal{H})$ can be specified by giving the number of particles in each state of $\mathcal{H}$:

$$|n_1, n_2, ...\rangle$$

Let us define operators $a_i^\dagger, a_i$ which create/annihilate a particle in state $|i\rangle$. These operators must satisfy

$$[a_i, a_j^\dagger] = \delta_{ij}, \qquad [a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0$$

An arbitrary state $|\psi\rangle \in \mathcal{F}$ is given by

$$|\psi\rangle = \sum_{n_1, n_2, ... \in \mathbb{Z}} c_{n_1, n_2, ...} |n_1, n_2, ...\rangle$$

where $|n_1, n_2, ...\rangle = \frac{(a_1^\dagger)^{n_1}}{\sqrt{n_1!}} \frac{(a_2^\dagger)^{n_2}}{\sqrt{n_2!}} ... |0\rangle$.

The operator $a_i^\dagger$ has no eigenstates: if $|\psi\rangle$ has a minimum of N particles, $a_i^\dagger|\psi\rangle$ necessarily has a minimum of $N+1$ particles. The operator a_i, however, has eigenstates: coherent states.

$$|\psi\rangle = \exp\big(\sum_i \psi_i a_i^\dagger\big)|0\rangle$$

we denote $\psi = (\psi_1, \psi_2, \psi_3, ...)$. For a free scalar field, for example, with a Hamiltonian $H = \sum_i \omega_i a_i^\dagger a_i$, the coherent state is simply a product of harmonic oscillator coherent states. We list two important properties of coherent states:

1. they form an overcomplete basis of $\mathcal{F}$:

$$\int d(\bar{\psi}, \psi) e^{-\bar{\psi}\cdot\psi} |\psi\rangle\langle\psi| = \mathbb{1}_{\mathcal{F}}$$

   where $d(\bar{\psi}, \psi) = \prod_i \frac{d\bar{\psi}_i d\psi_i}{\pi}$ and we denote $\bar{\psi} \cdot \psi = \sum_i \bar{\psi}_i \psi_i$. This is because the coherent states themselves are not orthogonal, so we need an extra exponential factor to get to unity.

2. $\langle\psi|\psi\rangle = e^{\bar{\psi}\cdot\psi}$

3.1.3 The Path Integral

In general, any system can be described by a partition function

$$Z = \mathrm{Tr}[e^{-\beta H}]$$

where H is the system's Hamiltonian, $\beta = \frac{1}{T}$, and we use units so that $k_B = 1$.

Important properties of the system can be calculated from Z (such as correlation functions and thermodynamic properties), and it can be evaluated as a field integral. Consider a normal ordered Hamiltonian $H(a^\dagger, a)$. We begin by inserting identities, using property "1" of coherent states:

$$\begin{aligned} Z &= \sum_n \langle n|e^{-\beta H}|n\rangle = \int d(\bar{\psi}, \psi) e^{-\bar{\psi}\cdot\psi} \sum_n \langle n|\psi\rangle\langle\psi|e^{-\beta H}|n\rangle \\ &= \int d(\bar{\psi}, \psi) e^{-\bar{\psi}\cdot\psi} \langle\psi|e^{-\beta H(a^\dagger, a)}|\psi\rangle \end{aligned}$$

where $\{|n\rangle\}$ forms a basis of $\mathcal{F}$. We cannot simply plug in ψ in place of the annihilation operators a because $e^{-\beta H}$ is *not* normal ordered. Thus we must resort to time-slicing: we divide the "time" $\beta = N\delta$ into N segments

$$Z = \int d(\bar{\psi}, \psi) e^{-\bar{\psi}\cdot\psi} \langle\psi|e^{-\delta H} \wedge e^{-\delta H} \wedge ... \wedge e^{-\delta H}|\psi\rangle$$

and insert new identities in place of every "$\wedge$", labelled by time slices $i = 0, ..., N-1$.

$$Z = \int d(\bar{\psi}, \psi) e^{-\bar{\psi}\cdot\psi} \Big(\prod_{i=1}^{N-1} d(\bar{\psi}^i, \psi^i) e^{-\bar{\psi}^i\cdot\psi^i} \langle\psi^{i+1}|e^{-\delta H}|\psi^i\rangle \Big) \langle\psi^1|e^{-\delta H}|\psi\rangle$$

Renaming $\psi \to \psi^0 = \psi^N$:

$$Z = \int \prod_{i=0}^{N-1} [d(\bar{\psi}^i, \psi^i) e^{-\bar{\psi}^i \cdot \psi^i} \langle \psi^{i+1} | e^{-\delta H} | \psi^i \rangle]$$

but $\langle \psi^{i+1} | e^{-\delta H(a^\dagger, a)} | \psi \rangle = \langle \psi^{i+1} | \psi^i \rangle e^{-\delta H(\bar{\psi}^{i+1}, \psi^i)} = \exp(\bar{\psi}^{i+1} \cdot \psi^i - \delta H(\bar{\psi}^{i+1}, \psi^i))$, where we used property "2" of coherent states:

$$\begin{aligned} Z &= \int \prod_{i=0}^{N-1} d(\bar{\psi}^i, \psi^i) \exp[(\bar{\psi}^{i+1} - \bar{\psi}^i) \cdot \psi^i - \delta H(\bar{\psi}^{i+1}, \psi^i)] \\ &= \int \Big(\prod_{i=0}^{N-1} d(\bar{\psi}^i, \psi^i) \Big) \exp \Big[\delta \sum_{k=0}^{N-1} \Big(\frac{\bar{\psi}^{i+1} - \bar{\psi}^i}{\delta} \cdot \psi^i - H(\bar{\psi}^{i+1}, \psi^i) \Big) \Big] \end{aligned}$$

Finally, taking the limit $N \to \infty, \delta \to 0$, the sum over time slices becomes an integral over time. Using $\int_0^\beta d\tau \partial_\tau \bar{\psi} \cdot \psi = -\int_0^\beta d\tau \bar{\psi} \cdot \partial_\tau \psi$, the partition function can be expressed as a *coherent state* field integral [5]:

$$Z = \int D(\bar{\psi}, \psi) \exp(-S[\bar{\psi}, \psi])$$

where $D(\bar{\psi}, \psi) = \lim_{N \to \infty} \prod_{i=0}^{N-1} d(\bar{\psi}^i, \psi^i)$. This measure expresses an integration over all values that a path can assume at every time slices. The integral is taken with limits $\psi(0) = \psi(\beta)$, and S is given by

$$S[\bar{\psi}, \psi] = \int_0^\beta d\tau (\bar{\psi} \cdot \partial_\tau \psi + H(\bar{\psi}, \psi)) \tag{4}$$

We can interpret this as an (Euclidean) classical Hamiltonian action, but instead of generalized coordinates and momenta, we use conjugate operators $(a, a^\dagger)$, so that $(\bar{\psi}, \psi)$ would by interpreted as "coordinates" specifying the system's state at any given time, much like (q, p).

Remembering the properties of coherent states, these new coordinates $(\bar{\psi}, \psi)$ describe a field with average particle number $n = \bar{\psi} \cdot \psi$. As the field evolves in time, the average particle count in each state, $n_k = \bar{\psi}_k \psi_k$ may change, yielding changes in the spatial profile of the field. Thus, the "paths" over which we integrate assume values in an "average number of particles" phase space $(\bar{\psi}, \psi)$.

For concreteness, for a scalar field the Hamiltonian is

$$H = \sum_i \omega_i a_i^\dagger a_i$$

The action (4) for this system is then

$$S[\bar{\psi}, \psi] = \int_0^\beta d\tau \sum_i \bar{\psi}_i(\tau)(\partial_\tau + \omega_i)\psi_i(\tau)$$

3.2 Phase Transition

As shown in [5], the Ginzburg-Landau model can be used as an effective theory for ferromagnetic systems in d dimensions at low temperatures. It can also be fully constructed from symmetry and phenomenological considerations [8]. The action is:

$$S[\phi] = \int d^d x \left[\frac{1}{2}(\nabla\phi)^2 + \frac{r}{2}\phi^2 + g\phi^4 - h\phi \right]$$

where $r = (1 - 2\beta C(0))$, $C(x-y)$ is a correlation matrix between sites, ϕ is the magnetization of the material, g is the coupling to a quartic interaction, $h = \beta B$ and B is an external magnetic field. The loss of permanent magnetization can be explicitly seen from studying this action. We find an estimate of Z by applying a mean field (saddle point) approximation. For this we must minimize the action. Because of the positive term involving a gradient, the mean-field configuration must be spatially constant.

$$\frac{\delta S}{\delta \phi} = 0 \iff r\bar{\phi} + 3g\bar{\phi}^3 - h = 0$$

Ignoring the external field h for a moment, we can see that for $r > 0$, the action will have a minimum at $\bar{\phi} = 0$ (see figure 2). This corresponds to a paramagnetic phase, with no net magnetization. Conversely, for $r < 0$ there are two minima: $\bar{\phi} = \pm\sqrt{|r|/3g}$. The system must choose between the two minima, that is, its symmetry is spontaneously broken and it enters a ferromagnetic phase, with permanent magnetization $\pm\sqrt{|r|/3g}$. If h is turned on, the degeneracy will be lifted, and the system will populate its lowest energy level.

Therefore the factor r is responsible for the phase transition, determining a critical temperature, beyond which the material becomes ferromagnetic or paramagnetic:

$$r = (1 - 2\beta C(0)) = \beta(T - 2C(0)) = \frac{T - T_C}{T}$$

T_C depends on the microscopic properties of the material.

There are other systems that present similar behaviour near the critical value of some parameter. For example, let's consider the (Euclidean) action of a weakly interacting Bose gas:

$$S[\psi] = \int d^d r \int_0^\beta d\tau \left[\bar{\psi}(\partial_\tau + H_0 - \mu)\psi + \frac{g}{2}(\bar{\psi}\psi)^2 \right]$$

where $H_0 = \hat{p}^2/2m$ is the free Hamiltonian, μ is the chemical potential, the fields are periodic in imaginary time $\psi(r,\beta) = \psi(r,0)$ and $0 < g \ll 1$.

Setting first $g = 0$, and assuming we are dealing with a system of size L, we can diagonalize the Hamiltonian by switching to momentum space:

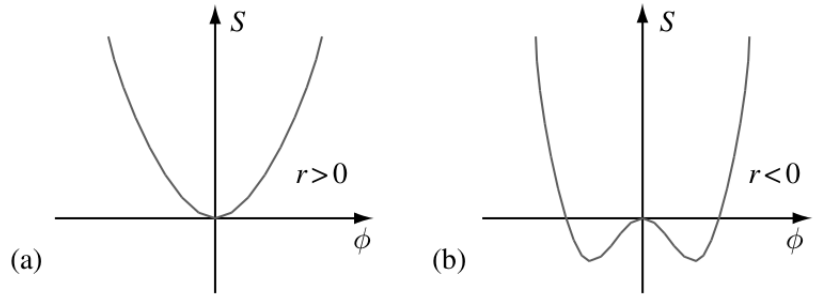


Figure 2: Value of the Ginzburg-Landau action S for a constant mean field configuration ϕ in the (a) paramagnetic and (b) ferromagnetic regimes. (Reproduced from [5]).

$$\psi(r) = \frac{1}{\sqrt{L^d}} \sum_a \psi_a e^{ik_a r}$$

so we get the following coherent state field integral:

$$Z_0 = \int D(\bar{\psi}, \psi) \exp\Big[- \int_0^\beta d\tau \sum_a \bar{\psi}_a(\tau)(\partial_\tau + \epsilon_a - \mu)\psi_a(\tau) \Big] \tag{5}$$

where $\epsilon_a = k_a^2/2m$ and it was used that $\int d^d r e^{-ir(k_a - k_{a'})} = L^d \delta_{aa'}$. We can make a transformation to Matsubara frequency representation:

$$\psi_n = \frac{1}{\sqrt{\beta}} \int_0^\beta d\tau \psi(\tau) e^{i\omega_n \tau}$$

where $\omega_n = 2n\pi T$, $n \in \mathbb{Z}$. The partition function takes the form

$$Z_0 = \int D(\bar{\psi}, \psi) \exp\Big[- \sum_{an} \bar{\psi}_{an}(-i\omega_n + \epsilon_a - \mu)\psi_{an} \Big] = \prod_{n,a} [-i\omega_n + \epsilon_a - \mu]^{-1}$$

where ω_n are Matsubara frequencies and ϵ_a are eigenstates of the free Hamiltonian. For bosons, the low-temperature excitations are around the vacuum state $|0\rangle$, and we can always set $\epsilon_0 = 0$ by adjusting μ.

We note that, for the action to be stable, $\mu \leq 0$.

The free energy is then given by

$$F = -T \ln \mathcal{Z}_0 = T \sum_{a,n} \ln(-i\omega_n + \epsilon_a - \mu)$$

where T is temperature. From this we can calculate the number of particles in the system as

$$N = -\frac{\partial F}{\partial \mu} = -T \sum_{n,a} \frac{1}{i\omega_n - \epsilon_a + \mu} = \sum_a n(\epsilon_a)$$

Here, $n(\epsilon) = (e^{\beta(\epsilon-\mu)} - 1)^{-1}$ is the Bose distribution. Taking a system with a fixed number of particles N, the equation above sets the dependence of μ on T: as T decreases, μ must increase up to its maximum value $\mu(T_C) = 0$. As μ approaches this point,

$$\sum_{a \neq 0} n_B(\epsilon_a, \mu \to 0) = N' \neq N$$

so that higher energy states cannot support enough particles - the ground state (with energy ϵ_0) accumulates a macroscopic number of particles $N - N'$. This phenomenon is known as Bose-Einstein condensation.

We can apply again a mean-field analysis and ask what is the behaviour of the system near its critical point $\mu = 0$. The relevant degree of freedom is the zero-mode (ground state) component of the field, $\psi_{a=0}$. Notice that the action for this mode is unbounded, because $\epsilon_0 = \mu = 0$.

This can be resolved if, inspired by the knowledge of the macroscopic behaviour of ψ_0 (remember that $|\psi_0| = N - N'$), we treat it as a *time-independent* Lagrange multiplier that fixes the number of particles in the condensate. Effectively, we are neglecting quantum fluctuations. Referring back to (5):

$$S_0[\psi] = -\bar{\psi}_0 \beta \mu \psi_0 + \sum_{a \neq 0, n} \bar{\psi}_{an}(-i\omega_n + \epsilon_a - \mu)\psi_{an}$$

where, as before, ψ_0 is the ground-state mode and ψ_{an} is the Matsubara component n of the field mode a. Now, when performing the integration over field configurations, only those with fixed $|\psi| = N - N'$ can be considered.

Now we may turn on the interactions by taking a small coupling g. We expect the ground state characterization as a spatially and temporally constant mode ψ_0 to survive and be macroscopically occupied below the condensation point. Moreover the action for the zero field will become

$$S = -\beta \mu \bar{\psi}_0 \psi_0 + \frac{\beta g}{2L^d} (\bar{\psi}_0 \psi_0)^2$$

Notice that, however small g may be, the second term above guarantees stability for the action (figure 3) , so that for the weakly interacting Bose gas, μ can take any value. The action of the zero mode is no longer unstable and we can treat it again as an integration variable, performing the mean-field analysis:

$$T \frac{\delta S}{\delta \phi} = \bar{\psi}_0 \Big(-\mu + \frac{g}{L^d} \bar{\psi}_0 \psi_0 \Big) = 0$$

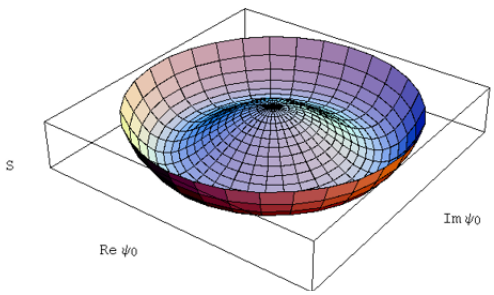


Figure 3: Graph of $S[\bar{\psi}_0, \psi_0]$, for $g > 0$ and $\mu > 0$.

For $\mu \leq 0$, above the condensation point, the solution must be 0. For $\mu > 0$, below the condensation point, this is solved by any ψ_0 such that $|\psi_0| = \sqrt{\mu L^d/g} = \gamma$, which, being proportional to the (square root of) the system's size, is evidently a macroscopically large value.

4 Conclusion

Now that we established the possibility of an analog of the VAE in condensed matter systems near phase transition, it remains to ask whether this analogy can be extended, or strengthened. For this we must examine the behaviour of these systems when transitioning from stable to unstable configurations (see Figure 2), searching for analogs to the tachyonic modes found in the VAE. This is the goal of this project for the next 6 months.

The study chronogram in table 1 is being followed:

Table 1: **Chronogram**

	Gen. Relativity	QFT	QFT in CM
2016 (2nd half)	(done)		
2017 (1st half)	(done)	(done)	(done)
2017 (2nd half)		(done)	(done)
2018 (1st half)			X

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